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CS 621 DATA ANALYTICS

Non-Numerical Data

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TYPES

- Categorical/Nominal
  - Color: Red, Blue, White, …
  - Major: CS, Math, Accouting
- Binary
  - True/False
  - Yes/No
  - Positive/Negative
- Ordinal/Ranking
  - Gold, Silver, Bronze
  - Freshman, Sophomore, Junior, Senior

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COUNT/FREQUENCY

- For each distinct value, compute its count/frequency.
- Commonly, we use a dictionary to store each distinct value as the key, and its frequency as the value.
- [‘B‘, ‘A’, ‘B’, ‘C’, ‘C’, ‘A’, ‘F’, ‘A’, ‘B’, ’A’]
- {‘B’:3, ‘A’:4, ‘C’:2, ‘F’:1}

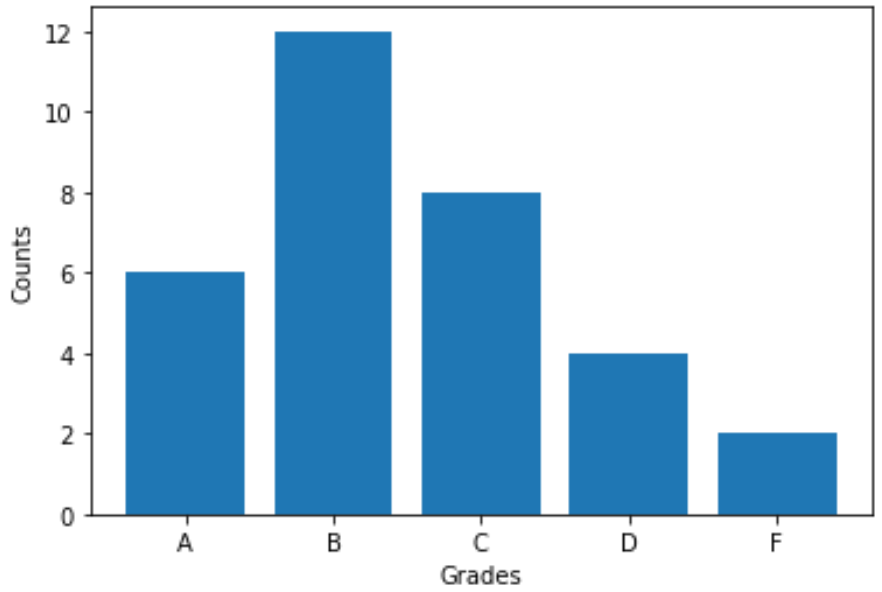
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BAR CHARTS

- For non-numerical data, Bar Charts are often used for visualization purpose.

- X = [‘A’, ‘B’, ‘C’, ‘D’, ‘F’]
- Y = [6, 12, 8, 4, 2]
- plt.bar(X, Y)

- plt.xlabel(“Grades”)
- plt.ylabel(“Counts”)



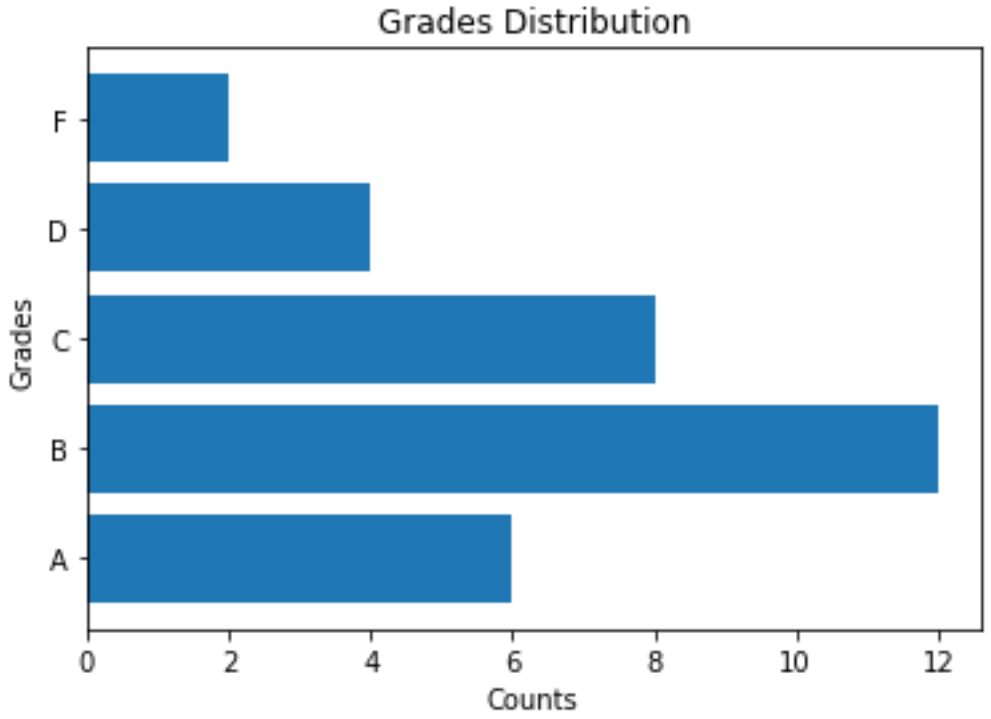
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HORIZONTAL BAR CHARTS

- Sometimes, bars were presented horizontally.

- X = [‘A’, ‘B’, ‘C’, ‘D’, ‘F’]
- Y = [6, 12, 8, 4, 2]
- plt.barh(X, Y)

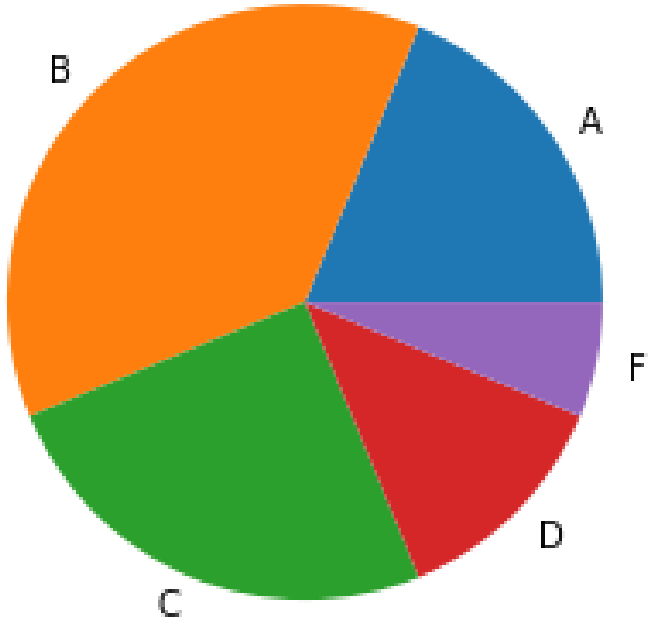
- plt.xlabel(“Counts”)
- plt.ylabel(“Grades”)



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PIE CHARTS

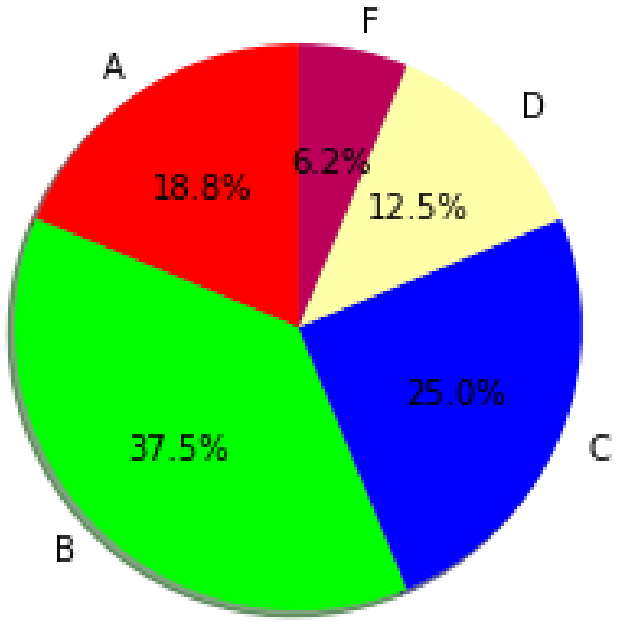
- For non-numerical data, Pie Charts are also used for visualization purpose.



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PIE CHARTS CONTD

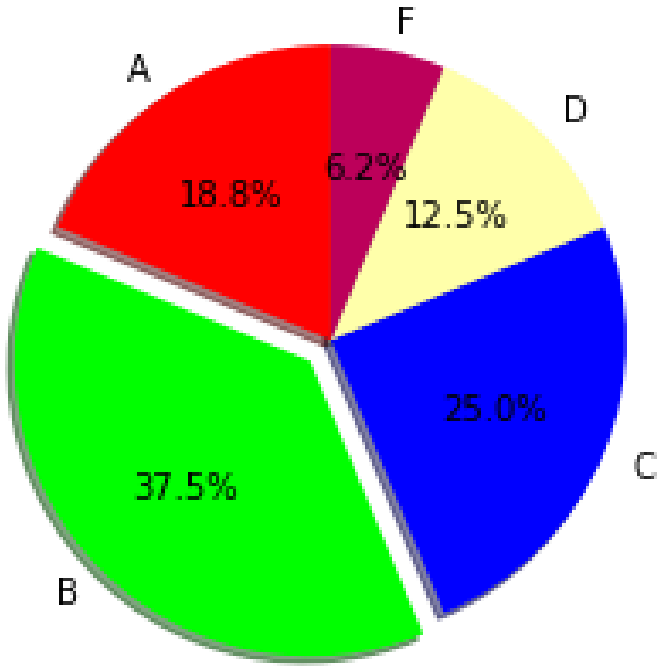
- One advantage of using Pie Chart is to show the percentage of each non-numerical value.



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PIE CHARTS CONTD

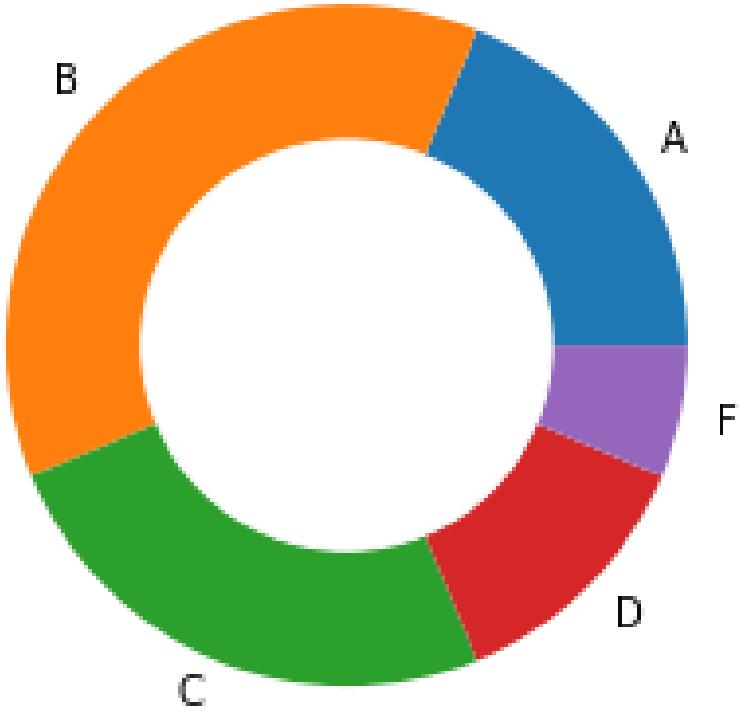
- Attention is on a particular slice of the pie.



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DONUT CHARTS

A Pie Charts with empty center is a Donut Chart.



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NUMERICAL AND NON-NUMERICAL DATA IN ONE TABLE

- A dataset often has both numerical and non-numerical values.
- Commonly, we will identify the “groups” on non-numerical data first, then apply numerical computation on the numerical values within each group.
- Pandas DataFrame has useful features for this kind of computation. Please see code demonstration for real examples.

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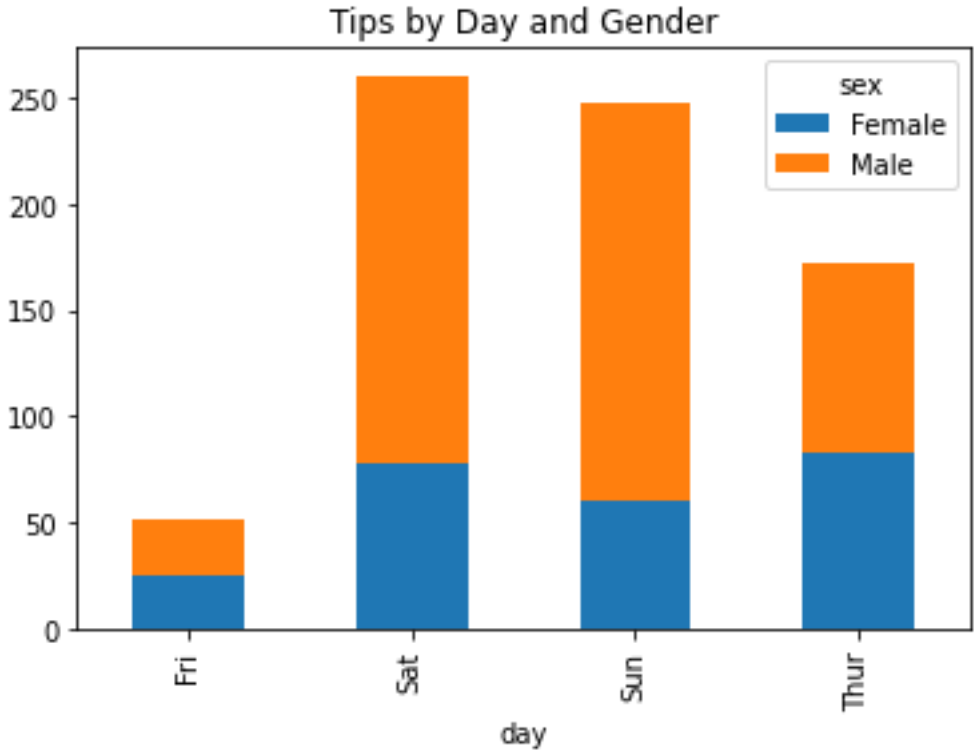
BAR CHART AND PIE CHART IN SERIES AND DATAFRAME

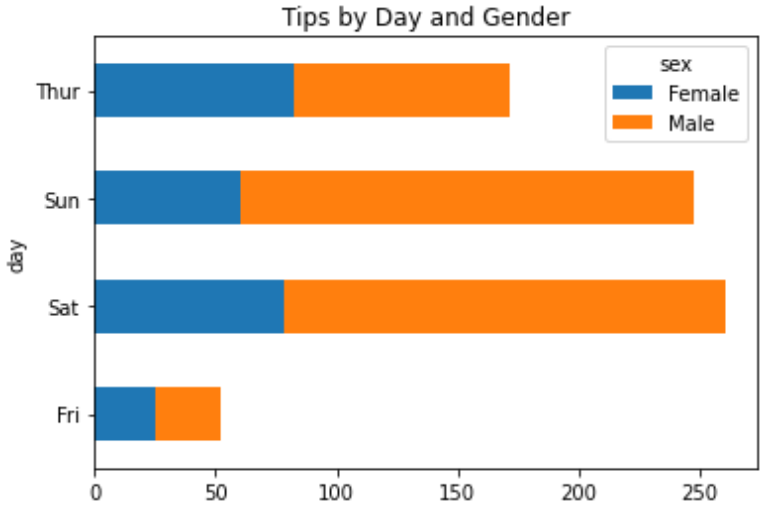
- Use df[‘categorical column].value_counts() to count each distinct value in a categorical column. Will return a Series object, e.g. ser.
- ser.plot.bar()
- ser.plot.pie()
- df.plot.bar(x = categorical column, y= numerical column)
- df.plot.pie(y= numerical column) # use index as labels

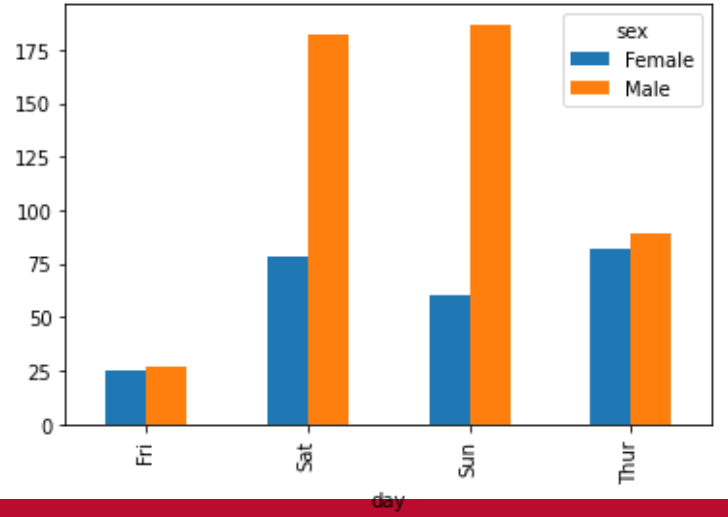
- df.groupby() and df.sort_values(by=[]) are useful for data mixed with non-numerical and numerical values.

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STACKED BAR CHARTS

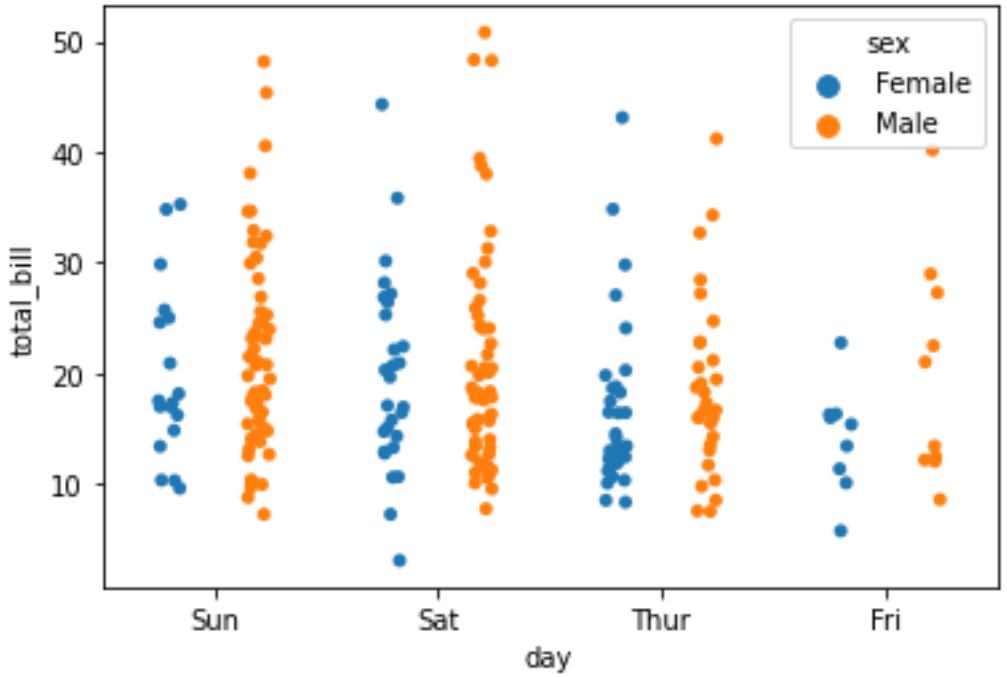


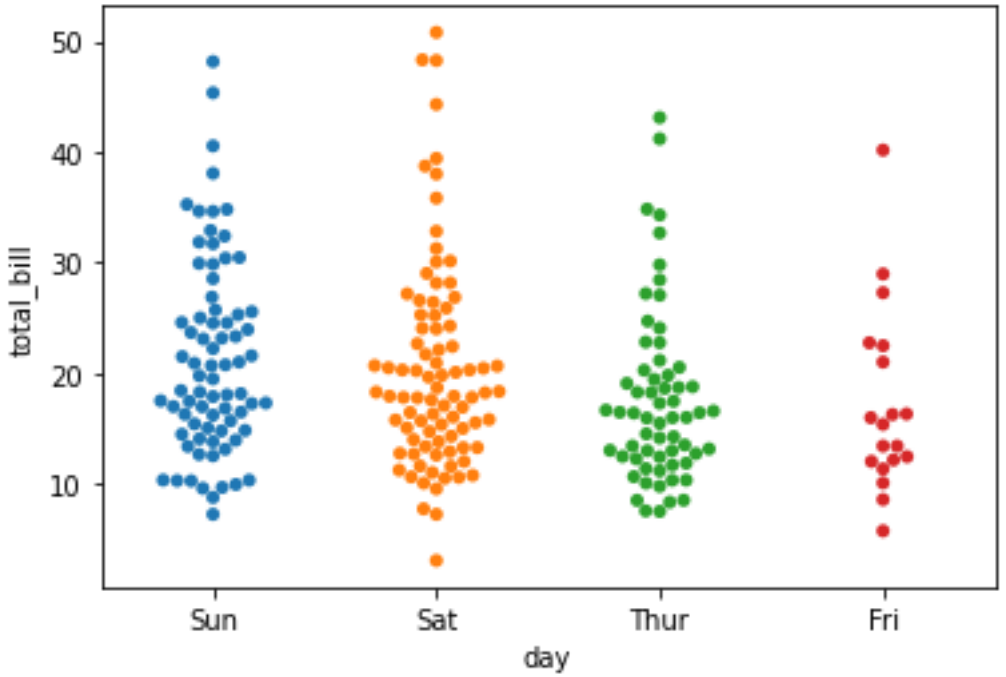




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STRIP AND SWARM PLOTS







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WE FLY